Supplementary Material for

Developmental Dilemmas

The Role of Power and Agency

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Appendix I
Political Settlements: Concept and Typology[1]

Political settlements shape developmental prospects. They both create and resolve collective-action problems.

Three related definitions: [2]

'A political settlement is an interdependent combination of a structure of power and institutions at the level of a society that is mutually 'compatible' and also 'sustainable' in terms of economic and political viability' (Khan, 2010, 20).

Political Settlements are mutual understandings, held among elites and powerful organizations, to use politics rather than organized violence as the primary means to settle internal disputes (Ferguson, 2020, 235).

A political settlement is 'an ongoing agreement among a society's most powerful groups over a set of political and economic institutions expected to generate for them a minimally acceptable level of benefits, which thereby ends or prevents generalized civil war and/or political and economic disorder' (Kelsall et. al, 2022, 46).

Characteristics:

- They are not necessarily written or formally negotiated.
- They are outcomes of history whose origins, operation, and longevity reflect distributions of power.
- Relevant distributions of power have a horizontal dimension across insider groups and a vertical dimension between elites and followers within groups.
- Political settlements are adaptive responses to changes in social environments.
- By structuring exercises of power, political settlements underlie the creation and trajectories of institutional systems.
- There operation reflects a punctuated-equilibrium dynamic with relatively long periods of stability followed by sudden bursts of rapid change that accompanies their collapse or reconfiguration.
- Political settlements create and reflect social equilibria whereby given institutional systems, however informal or disorganized, facilitate sufficient distribution of benefits to powerful agents—meaning at least minimal provision to deter such agents from undermining the settlement.

Classification of political settlements facilitates inquiry into relationships between distributions of power, the composition and motivation of political constituencies, configurations of institutions, distributions of benefits, collective-action problems, and developmental prospects.

Political settlements may be categorized along two dimensions, which are poles on spectra.

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[1] This appendix draws on Chapters 7-8 of Ferguson (2020) and Chapters 2-3 of Kelsall et. al. (2022).

[2] Related definitions: '. . . the informal and formal processes, agreements, and practices that help consolidate politics, rather than violence, as a means for dealing with disagreements about interests, ideas and the distribution and use of power' (Laws and Leftwich, 2014, 1)

Table A1: A Basic Typology of Political Settlements[3]		
	Configuration of Political Power	
Social Foundation	*Incoherent*	*Coherent*
Broad	Q1	Q2
Narrow	Q3	Q4

Vertical Dimension: The *social foundation* specifies the socially salient groups that are included in a settlement: its insiders. *Insiders* are groups to whom policy must respond—without coordinated repression—whether or not they belong to a governing coalition. *Social salience* requires either the presence of substantial numbers (say > 5% of the population) or, for smaller groups, the possession of notable disruptive power. Failure to accommodate (or coopt) insider groups in some, perhaps minimal, sense could motivate efforts to undermine the settlement. In other words, insiders must receive at least minimal side payment or benefit within a settlement to preclude resolving often substantial collective-action problems of actively disrupting it. Chinese government policy, for example, responds to needs of the rural populations, even though many have little official national representation. Certain foreign diplomats or officials from powerful NGOs or foreign corporations, though not representative of domestic populations, possess sufficient disruptive potential to become insiders. The social foundations of political settlements often has transnational elements.

Excluded groups, by contrast, do not receive side benefits. They can be ignored (e.g., many indigenous peoples in parts of Latin America) or repressed—directly with force; less directly with threats, divide-and-rule maneuvers; and/or via symbolic manipulation, such as appeals to patriotism. Divide-and-rule techniques (triadic power format4; see Chapter 2) and symbolic manipulation can silence excluded groups at relatively low cost.

Implications:
The breadth of a social foundation influences a ruling coalition's incentives to distribute benefits—broadly or narrowly.

Political settlement analysis can thus address transnational as well as national and subnational exercises of power.

Horizontal Dimension: The *configuration of political power* (COPP), involves those who most directly influence policy: insider elites and powerful organizations.[4] At one end of this spectrum, a *coherent* COPP signifies a coherent allocation of decision-making procedures and authority—one that reflects prior resolution of insider collective-action problems related to bridging social cleavages, resolving disputes, delegating broad authority, and achieving certain basic coordination. Coherent (unipolar) COPPs exhibit at least rough insider consensus regarding broad contours of policy, such as relationships between the

[3] Table A1 merges terminology on the horizontal and vertical dimensions from Ferguson (2020), Kelsall et. al. (2022) and Hickey and Sen (2024). Table A1's wording for the horizontal dimension appears in Hickey and Sen (2024). Ferguson calls the horizontal dimension the *Configuration of Authority*, which ranges from *Multipolar* to *Unipolar*. Kelsall et. al, call the horizontal dimension the *Configuration of Power*, which ranges *Dispersed to Concentrated.* Table A1's vertical dimension uses terminology employed by both Ferguson (2020) and Kelsall et. al. (2024). Hicky and Sen call this dimension the *Configuration of Social Power*. In all three, the range is form *Narrow* to *Broad.* The core idea of these closely related typologies appears in Kelsall and vom Hau (2019). Related typologies appear in Khan (2010), Levy (2014), Di John and Putzel (2009), Laws (2012), and Laws and Leftwich (2014).

[4] A more complex approach could include an embedded vertical dimension signifying elite relations to their various constituencies.

state and religion. At the opposite end, an *incoherent* (multipolar) COPP signifies no such coherence, often reflecting active social cleavages that foment conflicts among insider factions. Multipolarity implies scattered and uncoordinated authority dispersed across various elites, coalitions, and other centers of power, such as regional governments, powerful firms, militias, and local patronage networks.

Implications:
Reflecting the extent of prior resolution of insider CAPs, the COPP underlies the degree to which insider elites can formulate policy without continuous renegotiation among factions.

All else equal, societies with multipolar incoherent COPPs exhibit more sub-national diversity in political arrangements than do those with coherent COPPs.

The Four Quadrants: Each quadrant implies certain sets of developmental collective-action problems. For example, in Q1, broad/incoherent, insiders face strong incentives to distribute benefits broadly, but they lack the organizational coherence to pursue long-term economic developmental objectives. Necessary structural transformation—meaning a shift from low- to high-productivity activities requires investment in human capital, technology, infrastructure and diversification of production towards high-productivity sectors and away form a focus on point-source resources and easily exportable commodities (Rodrik, 2013; Lin, 2012).[5] In Q1, for example, different insider elites, pursuing often discordant objectives, coopt groups in distinct segments of the broad social foundation (e.g., distinct regions, ethnicities) by offering short term benefits—often in the form of patronage for supporters and specific ethnic groups. Developmental CAPs, thus include both dilemmas regarding establishing coherent authority over political procedure and the allocation of benefits, without narrowing the social foundation and economic dilemmas regarding motivating long-term investments in productive structural transformation.[6]

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[5] For a typology of structural transformation and discussion of economic roots, see Sen (2023). See also Sen (2019) and Rodrik (2016).

[6] Ferguson (2020) elaborates on collective-action problems faced in the four quadrants.

Lin, J. Y. (2012) *New Structural Economics: A Framework for Rethinking Development and Policy.* Washington, D.C.: World Bank.

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Appendix 2: Leader-Follower Signaling Game

(This game illustrates principles from Sections 3 and 4.)

There are two players, a leader (L) and a potential follower (F). Prior history (nature) determines whether L is Tough (T) T or Weak (W).[7] On this basis, L decides whether to issue a command, which operates as a signal that L is type T. F, who does not know L's history, decides whether to obey or disobey. The consequences of disobedience depend on whether L is T or W.

Will F obey? Figure A2-1 shows a signaling game that illustrates the tradeoffs. The key question: Is there a perfect Bayesian separating equilibrium in which F chooses to obey?

Terminology:
A *separating equilibrium* requires that action (signal) reveals type (here, T or W for L).

In a *perfect Bayesian equilibrium*, each player chooses their best action at each information set, in light of its available information, and all players draw correct inferences from their observations, as specified by Bayes' rule for updating probabilities.

An *information set* is a collection of nodes with two properties: (i) one player has the move at all nodes in the set, and (ii) when the game reaches any node in the information set, the player does not know which node has been reached.[8]

The following details apply to this game:

- L signals that its type is T by issuing a command. If L does not issue a command, L does not signal anything (issues no signal).
- L, whether T or W, always prefers that F obey a command, and L prefers an obeyed command to no action.
- Signaling (i.e., issuing a command) is costly for leaders of both types T and W.
- If F disobeys a signal, conflict follows. Conflict is costly to both L and F (both lose from a fight). Conflict is more costly to both players if L is type T—as in a prolonged fight.

The following three questions arise:

1. Given the cost to L of signaling and the greater costs of conflict to both players if L is T, would it be worthwhile for a type-T and only a type-T leader to signal by issuing a command?
2. If F receives a signal, will F believe that L is really type T?
3. If so, will F choose to obey?

Additionally, to solve the game, we must specify how F's updates beliefs after receiving a signal (S) or no signal (NS). To address the possibility of a separating equilibrium, we assume that F would interpret S as only coming from a T leader and interpret no signal (NS) as only coming

[7] One could devise a pre-game to determine whether L is T or W.
[8] This label may appear slightly ironic, but an information set does specify information available to the relevant player.

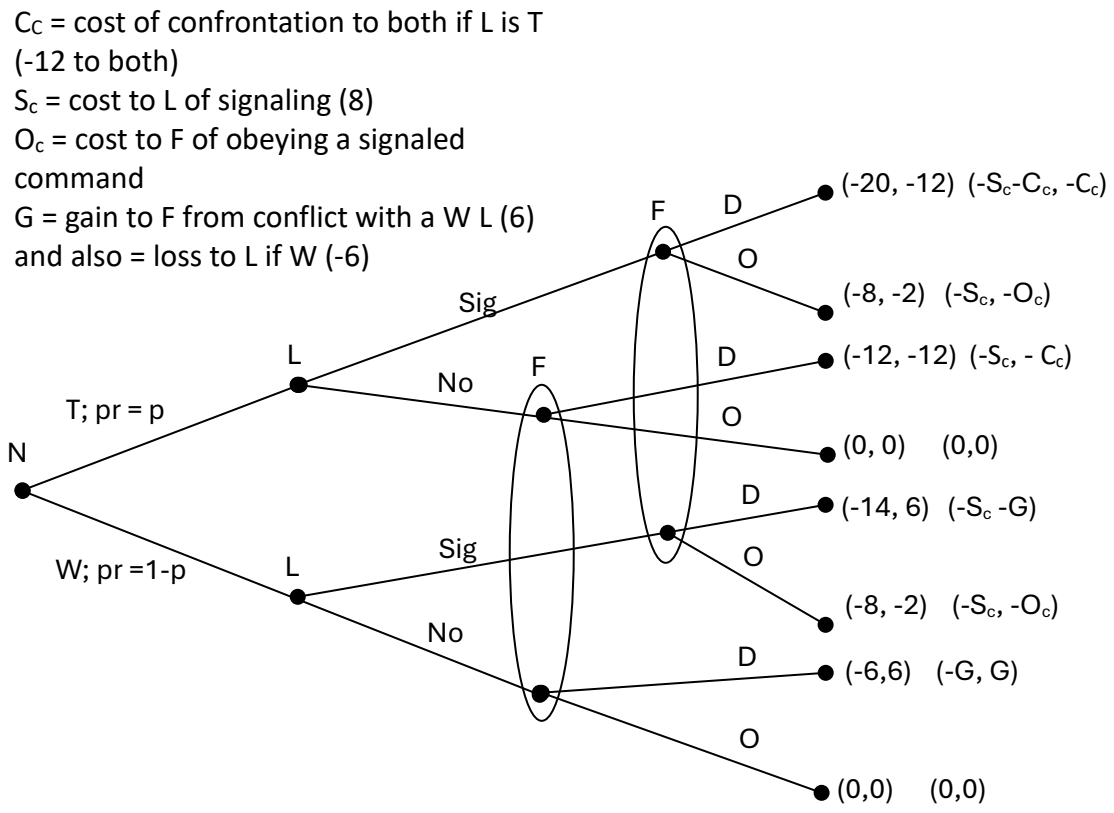
from a W leader.[9] Note this assumption does not determine the game's outcome, but only the possibility of achieving a separating equilibrium. Questions 1-3 remain. Given this updating assumption, does the configuration of payoffs allow for a perfect Bayesian separating equilibrium?

The Sequence of Moves[10]

1. During period t_0, 'nature' (N), reflecting prior history (possibly the outcome of a pre-game), determines whether L is Tough (T) or Weak (W). L observes N's move, but F does not.

2. In period t_1, L decides whether to signal a command (S) or not signal (NS). Signaling comes at cost S_c. The signaled command could either be a simple directive, 'do not enter', or a general instruction ('do as I say'). In the latter case, L acts as a 'focal person' (Basu, 2021), as illustrated in Chapter 3's discussion of Prigozhin and the Wagner Group.

Figure A2-1: Leader-Follower Signaling Game



[9] This could reflect F's predisposition or the outcome of prior encounters with L or people who resemble L.
[10] Figure A2-1 closely resembles Figure 9.6 in Dixit and Skeath (2004, 283).

3. In period t_2: F, having not observed N's move, observes L's move (S or NS) and then, not knowing whether L is T or W, decides whether to obey (O) or disobey (D). The relevant information sets appear as circles in the diagram.

4. Payoffs are awarded as shown with a numerical example and with symbols (defined in the diagram). The first payoff goes to L; the second to F.

Returning to the questions:
Given F's updating inference and the shown payoffs, is there a perfect Bayesian separating equilibrium?[11]

With the payoffs shown, the game can depict a perfect Bayesian separating equilibrium. Given the updating assumption, there are four steps to the backward induction reasoning.

a. If F receives S, will it obey? Yes. F interprets S as coming form T. Since -2 > -12 ($C_c > O_c$) F chooses to obey.

b. If F receives NS will it disobey? Yes. F interprets S as coming from W. Since 6 > 0 (G > 0) F chooses to disobey.

Using backward induction, L can anticipate possible outcomes a. and b.

c. Will a T leader want to signal? Yes, because the cost of signaling is less than the cost of both signaling and fighting: -8 > -20 ($C_c > 0$). (If L were to choose NS, F would think leader is W and would disobey, leading to -20.)

d. Will a W leader choose NS? Yes, because loss from conflict is less than the cost of signaling: -6 > -8 ($C_c > G$).

Thus, we have a separating perfect Bayesian equilibrium. Actions S and NS signifies leader's type, T or L. At every information set, F is offering its best response, given its information and its updating assumption. Likewise, L uses backward induction to generate its best response to the anticipated actions of F. Moreover, F's updating inference (that S means T and NS means W) is shown to be correct.

More generally, this outcome will obtain, for the given inference, as long as all the variable payoff inequalities shown in items 1-4 hold.

If, for example, $C_c < G$, a W leader would also signal. If so, F's updating inference would be incorrect. There would be no separating perfect Bayesian equilibrium.

This game can thus illustrate a key dynamic within Chapter 3's discussion of leader-follower dynamics as illustrated by Prigozhin and the Wagner Group, with Prigozhin as a Tough leader.

[11] Figure A3-1 resembles Figure 9.6 in Dixit and Skeath (2004, 283).

Figure A2-1, with a slightly different definition of signaling can also represent Chapter 4's discussion of introducing new norms. Here, consider leader L as an institutional entrepreneur (IE), and define signal (S) as L's transmission of a new behavioural prescription: a new ADIC statement (see Chapter 4), with the intent of instigating a new social norm. Here, 'Obey' connotes adhering to the new normative prescription. The cost of (or loss from) disobeying (C_c or -G as relevant) now signifies the combined influence of guilt, anger, and shame faced by non-adherents, noting that guilt only applies to norm internalizes who nonetheless disobey (see equations (4) – (6) in Chapter 4. Furthermore, the anticipated cost of anger and shaming likely follows the conformity dynamics represented in Chapter 3's Figure 4 and a similar figure in Appendix 4.

We could complicate the model by also specifying whether L is prototypical or not (See Chapter 3). L's anticipated cost of signaling (S_c) is likely to be lower for a prototypical leader and may also decline as the anticipated number of adherents increases (again, an application of Figure 3). But the costs to conflict for both type T and type W could be higher for a prototypical L, who may face potential loss of a privileged position—perhaps especially if a prototypical L is type W. In such a case, we might assign different absolute values to G_F and G_L (a minor complication that could apply in Figure A2-1 as well). Additional variations could be developed.

Furthermore, a more sophisticated model could develop functions for each of the payoffs (S_c, C_c. O_c, and G), all of which may depend on distributions of power and other factors.

More complicated models could specify a series of related games in sequence. A two-stage game could first show Chapter 3's Figure 3 (or Figure A3-1) with payoff functions that reflect elements of Chapters 3 and 4. The outcome of that game would then influence the values for variables S_c, C_c. O_c, and G.

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Appendix 3: Model of IE Influence on Existing Norms

This model illustrates the influences of social norms on material and social payoffs. The latter represent praise, anger, and shame.

Consider individual *i*'s payoff from following (or adhering to) norm Ψ (strategy F) or violating Ψ (strategy V):[12]

(A3-1) $\omega_i(\Psi) = \pi_i(s_i, s_{-i}) + \beta(\eta_\Psi(s_{-i}(\iota_{-i})) - \xi_i(1 - \eta_\Psi(s_{-i}) + \partial\omega_i/\partial\iota_i \,|s_i = F)$, and
(A3-2) $\omega_i(\Psi) = \pi_i(s_i, s_{-i}) - g(\iota_i) - \varsigma(\iota_i; \eta_\Psi(s_{-i}(\iota_{-i})); \xi_\eta(\iota_\eta;(1 - \eta_\Psi(s_{-i})) \,|s_i = V)$.

Payoff ω_i is a weighted combination of material payoffs (π) plus social payoffs (the other terms); s_i and s_{-i} respectively signify strategies of *i* and other agents.

Equation (4-1) applies when *i* chooses strategy F (adhere or follow). The material payoff, π, is straightforward. β (praise or social approval from choosing F) increases payoff ω_i. β depends on proportional adherence to Ψ's prescription (η_Ψ), which depends on others' strategies ($\eta_\Psi(s_{-i}(\iota_{-i}))$. Additionally, follower *i*'s anger at violators (ξ_i) reduces ω_i. ξ_i depends on the proportion of non-adherents ($1- \eta_\Psi(s_{-i}(\iota_{-i}))$. Finally, i's internalization (ι_i; $0 \leq \iota_i \leq 1$) directly increases ω_i ($\partial\omega_i/\partial\iota_i > 0$).

Equation (4-2) offers similar logic when *i* chooses strategy V. The payoff loss from guilt (–*g*) depends on *i*'s internalization (ι_i). Payoff loss from shame (–ς) depends on ι_i and the proportion who choose F ($\eta_\Psi(s_{-i})$). Additional loss accompanies followers' expression of anger at *i*'s violation (ξ_η). ξ_η, depends on group internalization (ι_n) and the proportion of adherents ($\eta_\Psi(s_{-i})$).

Recall, Chapter 4's equation (4):

(4) $\Psi: h(A, C; N) \rightarrow D_N(I \,|A, C)$

Here, *h* is an information set: a shared mental model that conveys understanding of relevant social context in terms of A- and C- statements and N members of the relevant group. These terms jointly specify Ψ's scope: it applies to certain members of N (vai A), under specific circumstances (via C). Reflecting Ψ's strength, *h* then implies deontic (D_N): members of N 'should' undertake actions consistent with Ψ's intent (I), given A and C.

Further, recall the ADICO syntax (Crawford and Ostrom) as summarized in Chapter 4:

[12] A more detailed version of this model appears in Ferguson (2013).

- A = Attributes: the type of individuals to whom the norm applies, such as adult members of a specific ethnic group, women, or contract employees.
- D = Deontic: an ethical 'should'. One should contribute to community projects.
- I = AIM (middle letter), i.e., intent: the prescribed behaviour, such as 'do not cut in line' or 'respect your elders'.
- C = Conditions: the conditions under which the behavioural prescription applies, such as when addressing an elder or while attending a community meeting.

The arguments of (4-1) and (4-2) illustrate how components of *i*'s payoffs respond to (3)'s ADIC statements. Strategies (s_j and s_{-1}) depend on the shared mental model conveyed by information set *h*(A,C; N), which specifies Ψ's *scope* (it's A and C statements) within group N; *h* conveys Ψ's *intent* (I-statement) and *strength* (D-statement, D_N). The I-statement distinguishes an F-action from a V-action, indicating whether (4-1) or (4-2) applies. Anticipated consequences—the praise, anger, guilt, and shame conveyed by β, ξ, *g*, and ς, apply according to Ψ's A- C- and I- statements, as influenced by strength (D), and corresponding internalization levels ι_i and ι_n. $\partial(\cdot)/\partial\iota > 0$, and $\partial(\cdot)/\partial D > 0$.

Equations (4-1) and (4-2) offer detail that underlies Chapter 4's equation (4), which shows agent *i*'s strategy function.

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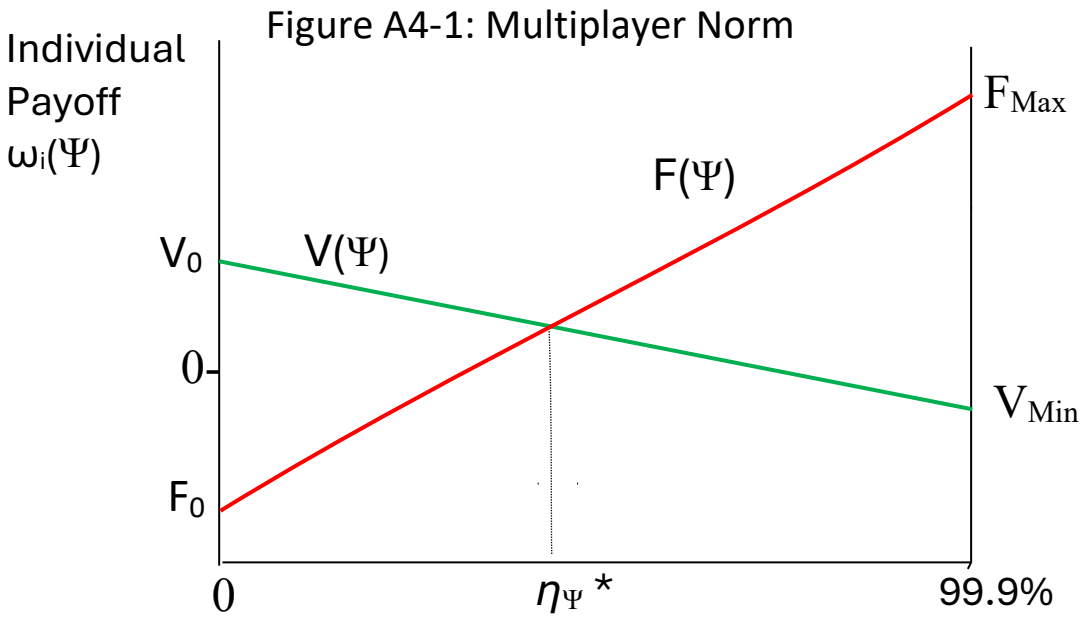
Ferguson, W. D. (2013) *Collective Action and Exchange: A Game-Theoretic Approach to Political Economy*. Stanford, CA: Stanford University Press.

Appendix 4: Chapter 3's Figure 4 Adjusted for Adherence to a New Social Norm Ψ



Legend: $\omega_i(\Psi)$ = *weighted combination of player* i's *material and*

For simplicity, we assume that all members of N have homogenous preferences regarding actions F and V. Individual *i*, a member of group N, decides whether to adhere to social norm (FΨ) or violate (VΨ). Payoff ω_i depends on the proportion of N who choose F (η_Ψ) as shown on both curves. Below (above) tipping point η_Ψ*, V(Ψ) shows the higher (lower) payoff: there are two stable Nash equilibria at $\eta_\Psi = 0$ and η_Ψ = N. Consequently, if an initial equilibrium rests at $\eta_\Psi = 0$, an institutional entrepreneur (IE, see Chapter 4) who desires to introduce a new norm or gain adherents to a controversial norm, faces a collective-action problem of generating sufficient adherence to cross the η_Ψ* threshold.

Appendix 5: The Emergence of Resource Conflict: Logic and Games

Individual members of group X understand the pluralistic and singular elements of their social identities. Here is Chapter 4's individual social identity equation:

(7) $I_i = \lambda_i E_i + (1-\lambda_i)Z_i$.

The two weighted terms, E and Z, respectively represent categorical ethnic/racial identity and Z represents other social classifications as well as the relational elements of identity.[13] λ represents the weight placed on E: its salience.

E's salience (λ_i) may be fixed in the short run but, over time, can respond to the following factors: the strength of group affiliations (τJ_i, where $J \in \{X, Y\}$ and τ = strength of affiliation); related social classifications of all individuals C_k, which is a function of relevant groups (J_k) and individual characteristics (ε_k); pertinent social norms (Ψ_{CJ}); resource conflict between groups X and Y (Λ_{XY}), a function of relative resource access ($R_X - R_Y$); and unequally weighted individual strategies and social identities (S_k and I_k), where weights reflect prestige and power. We have:

(8) $\lambda_i = \lambda_i (\tau J_i, \tau J_{-i}; C_k(J_k, \varepsilon_k), \Psi_{CJ}; \Lambda_{XY}(R_X - R_Y; S_k, I_k))$.

Members of group X are distributed along a spectrum of identity singularity; $0 \leq \lambda_{ix} \leq 1$, for all *i*. Members of rival group Y face similar identity functions. As in Chapter 4, group social identities are a weighted sum of individual identities, with unequal weights according to the internal distribution of power. Group social identity thus reflects the collective relative salience of E and Z. We have:

(9) $I_J = \Gamma_J E_J + (1- \Gamma_J)Z_J$.

Group singularity term, Γ_J, the unequally weighted sum of individual λs, is the variable of interest. Here is a simplified version of Chapter 4's equation (10) that focuses on group X, norms, proportionate adherence, and the distributional conflict over relative group access to contestable resource R_T (R_{TXY}):

(10') $\Gamma_X = \Gamma_X(,(\Psi_X, \eta_{\Psi X}; \Lambda_{XY}(R_{TXY})$.

Ψ_X = X's group norm; η_Ψ = the proportion of X that adheres to Ψ_X.[14]

Again, group Y has a similar group identity function.

Status-quo arrangements, supported by external authorities, allow both groups partial access to contestable resource R_T. Both have economic and cultural reasons for desiring full control, but challenging the status quo is risky.

Simmering conflict over R_T creates (and reflects) social cleavage, which IEs may exploit. The extent of X-Y social cleavage (Θ_{XY}) depends on a combined, weighted singularity term (Γ_{XY}). Each group's weight depends on group singularity (Γ_J) and common perceptions of X-Y conflict. IE and extremist perceptions

[13] See Chapter 4 for discussion of singular and pluralistic identities and categorical and relational components of social identity and for more details on equations (7) – (9).
[14] Group norms emerge from and reinforce strong group social identities. Chapter 4 elaborates.

play disproportionate roles (not modeled), as do actions of IEs and extremists (a_{EXTJ} and a_{IEJ}). Here is a slightly modified version of Chapter 4's equation (11):

(11') $\Theta_{XY} = \Theta_{XY}(\Gamma_{XY}, a_{EXTJ}; a_{IEJ})$.

$\partial\Theta_{XY}/\partial\Gamma_{XY} > 0$; $\partial\Theta_{XY}/\partial a_{ExT\,J} > 0$; $\partial\Theta_{XY}/\partial a_{IEJ} > 0$; and $\partial^2\Theta_{XY}/\partial^2\Gamma_{XY} > 0$.

The extent of X-Y social cleavage emerges (Θ_{XY}) depends on the combined singularity of group social identities (Γ_{XY}) and the actions of extremists and IEs. The impact of combined singularity on Θ_{XY} increases exponentially.

On this basis, the following two games represent the stages of Chapter 5's initiation of resource conflict. The first game illustrates the possible emergence of sharp social cleavage, as discussed in Chapter 5's stages 1-3. The second game represents the stage 2-3 transition from sharp social cleavage to possible inter-group fighting over resources.

First Game: The possible emergence of sharp social cleavage.

Consider the following game sequence, from the point of view of group X:

1. In Chapter 4's stage 1 period t_1, a group-X IE decides whether to invest, at cost c_M, in discovering and spreading political-economic and normative-identity memes to enhance X-Y social cleavages by augmenting Γ_{XY} via its determinates from Chapter 4's equation (10). A successful meme then offers extremists reputation benefit (Υ_{IE}) from initiating an incident (an attack on some member or members of Y). Υ_{IE} depends on X-Y social cleavage (Θ_{XY} from Chapter 4's (11), which depends on identity singularity (Γ_{XY}). Abstracting from other reputational influences, we have:

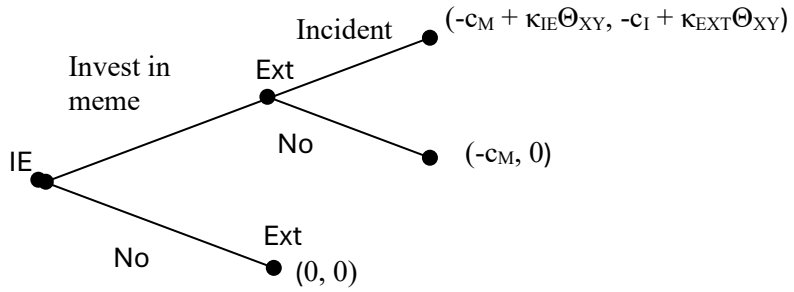
   (A5-1) $\Upsilon_{IE} = \kappa_{IE}(\Theta_{XY}((\Gamma_{XY}))$.
   κ_{IE} is a parameter. $\partial\Theta_{XY}/\partial\Upsilon_{IE} > 0$

   If undertaken, the meme enhances the salience of social identity singularity (Γ_X) by reinterpreting group social norm (Ψ_X), and corresponding ADIC statements (see Chapter 4) and by reshaping understandings of resource conflict (Λ_{XY}), moving towards a zero-sum interpretation. X's proportion of adherents ($\eta_{\Psi X}$) increases, especially among extremists—moving them towards a tipping point like that in Appendix 4's Figure 4 (which is a variation on Chapter 3's Figure 4). The probability of sharp social cleavage (Θ_{XY}) thereby increases. For extremists, these changes carry more salience than for average group members. Extremists experience larger impact on identity singularity (Γ_{EXTX}), which moves towards a tipping point for action (Γ_{EXTX}*; a variation on Figure 4 could illustrate), after which their version of (11)'s Θ_{XY} (Θ_{XYEXT}) would rapidly increase.

2. In stage 2's period t_2, following an additional round of IE memes (not shown in Game 1), Γ_{EXTX} passes tipping point Γ_{EXT}*. Extremist understanding of sharp social cleavage and its corresponding social cleavage, Θ_{XY}, rapidly escalates. Extremists then decide whether to execute a provocative incident, such as sabotage of a Y facility or an attack on a few prominent or perhaps random members of Y. Orchestrating an incident costs extremists amount c_I and confers reputation gain $\kappa_{EXT}\Theta_{XY}$.

   Game 1 represents the IE and extremist decisions in states 1 and 2. The tradeoff between players' reputational payoffs and costs of investing in memes and incidents determines the backward-induction outcome.

Game 1: IE and Extremist



Legend: Ext = extremist. c_M = IE's cost of signaling a meme; c_I = Extremist's cost of creating an incident. Θ_{XY} = X-Y social cleavage. Products $\kappa_{IE}\Theta_{XY}$ (=Υ_{IE}) and $\kappa_{EXT}\Theta_{XY}$ (= Υ_{IE}) show the respective reputation gains from social cleavage and conflict; κ_{IE} and κ_{EXT} are parameters.

A backward-induction conflict outcome (Invest, Incident) depends on two conditions:

i. If $\kappa_{EXT}\Theta_{XY} > c_I$, the extremist would choose Incident if IE chooses Invest.

ii. If so and if $\kappa_{IE}\Theta_{XY} > c_M$, IE would Invest in creating a conflict meme.

Hence, for given c_M and c_I, this outcome depends on the sensitivity of Θ_{XY} to the arguments of (11'): Γ_{XY}, a_{EXTJ}, and a_{IEJ}, and on underling determinates of Γ_{XY} from (10'). A more complicated model could show the relevant functions, whose arguments would influence the value of Θ_{XY} (perhaps as outcomes of a pre-game). Additionally, costs c_M and c_I, depend on the degree to which the IE and extremist are prototypical leaders (see Chapter 3) and their access to sources of power—that is, access to resources, ability to resolve organizational collective-action problems, and institutional or network positions. More detailed models could specify relevant functions for costs c_M and c_I (again, possibly as outcomes of pre-games).

Game 2: Group X-Y Resource Conflict

Background:[15]

Section 1 of Chapter 5 discusses three steps in the emergence of resource conflict. It offers a concluding equation (12) representing the ensuing intensity of conflict, and refers the reader to this appendix for a model of prior developments.

Consider the following: If Game 1 generates a stage-2 backward-induction outcome of (Invest, Incident), we proceed to stage 3.

In stage 3's period t_3, entrepreneurs, from both groups, now accompanied by extremists—several of whom become leaders and brokers (see Chapter 3)—invest in additional memes and incidents intended to recruit average group members to fight. On this basis, group members decide whether to invest amount *f* into fighting.

[15] The equations related to fighting utilize logic that appears in Shayo (2009) and Sambinis and Shayo (1023).

Now, consider the payoff to a member of group X from 'fighting' group Y over access to R_T.[16]

(A5-1) $\omega_{iX}(f_{iX}) = \pi_{iX}(f_i) + \Upsilon_{iX}(\Theta_{XYt-1}) = [q_i - c_f f_i + (1/N_X)\rho_{sX} R_T] + \Upsilon_{iX}(\Theta_{XY})$

Here, $\omega_{iX}(f_{iX})$ sums material payoff π_{iX} and reputational payoff $\Upsilon_{iX}(\Theta_{XY})$, an exponentially increasing function of social cleavage (Θ_{XY}); $\partial\Upsilon_{iX}/\partial\Theta_{XY} > 0$; $\partial^2\Upsilon_{iX}/\partial^2\Theta_{XY} > 0$. q_i = initial endowment; the product $c_f f_i$ = *i*'s investment in fighting resources; c_f = per unit cost of fighting capability, f_i; N_X = number of members in X; the product $(1/N_X)\rho_{sX}R_T$ = player *i's* share of the expected payoff to seizing R_T; ρ_{sX} = probability of successful seizure by X.[17]

Moreover, ρ_{sX} depends on X's share of total X and Y fighting resources ($F_X + F_Y$). For each, F depends on the group proportion who fight (η_{fJ}). ρ_{sX} also depends on resources that authorities and external actors devote to preserving the status quo (F_G). For X's probability of success, we have:

(A5-2) $\rho_{sX} = \rho_{sX}([F_X(\eta_{fX})/[(F_X(\eta_{fX})+F_Y(\eta_{fX})]; F_G)$.

For simplicity, let total fighting resources for either group J, $F_J(\eta_{fJ}) = \Sigma f_{iJ} = m_J f$, where η_{fJ} = the proportion of members of J who contribute amount *f* to fighting, and m_J = number of members of group J who contribute. Further, define X's portion of total fighting resources as $M_X \equiv m_X f_X/(m_X f_X + m_Y f_Y)$.

For simplicity we ignore F_G (which would shift the relevant functions), and simplify (A5-2) to

(A5-2') $\rho_{sX} = \theta M_X R_T$.
θ is a parameter; $\partial\rho_{sX}/\partial M_X > 0$.

For additional simplicity, assume a group J member's reputational payoff to fighting ($\Upsilon_{iX}(\Theta_{XY}) \equiv \kappa_{iJ}\Theta_{XY}$) depends only on the group proportion who choose to fight (η_{fJ}). Here, η_{fJ} represents social cleavage. Thus, for a member of X, we have:

(A5-3) $\Upsilon_{iX}(\Theta_{XY}) = \kappa_{iX}\Theta_{XY} = \sigma\eta_{fX}$.
$\sigma > 0$, is a parameter. $\partial\Upsilon_{iX}/\eta_{fX}\partial > 0$.

With this background and ignoring q_i, we can rewrite player *i*'s payoff from fighting (A5-1) as:

(A5-4) $\omega_{iX}(f_{iX}) = -c_f f_i + (1/N_X)\theta_i M_{XY} R_T + \sigma\eta_{fX}$.

Now, consider a similar function for the subset of group members who are extremists.

(A5-4') $\omega_{ExX}(f_{ExX}) = -c_f f_i + \theta_{EX} M_{XY} R_T + \delta\eta_{fX}$.

For both, parameter $\theta > 0$.

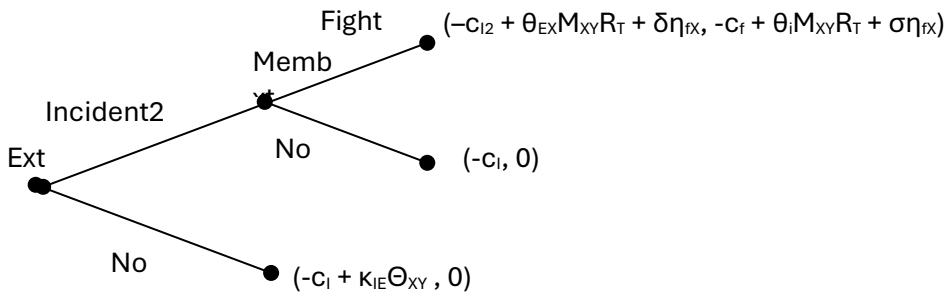
On this basis, if Game 1 generates outcome (Invest, Incident), we proceed to Game 2, which represents the stage 2-3 interaction between an extremist and an average group member.

Game 2

[16] As in Chapter 5, *fighting* may connote strikes, mass demonstrations, or violence. This discussion draws on modeling and discussion from Shayo (2009).
[17] A more complicated model could make c_f a separate variable.

Here, extremists decide whether to conduct a second incident, at cost c_{I2} with expected material benefit, $\theta_{EX}M_{XY}R_T$, and reputation benefit $\delta\eta_{fX}$. An average member then decides whether to fight. Considering the tradeoffs in (A5-1), (A5-3), (A5-4), and A5-4') we have:

Game 2: Extremist and Average Member Incidents and Fighting



Legend: c_{I2} = Extremist's cost of creating a second incident. Other terms remain as previously defined.

Given an (Invest, Incident) outcome from Game 1, there are two additional conditions for an (Incident, Fight) outcome.

iii. If Extremist chooses Incident2, Member *i* will Fight as long as the sum of material and social payoffs from *f* exceed c_f; if $\theta_i M_{XY}R_T + \sigma\eta_{fX} > c_f$.

iv. If so, with backward induction, Extremist conducts a second incident only if the reputation gain exceeds the dost of creating a second incident: $\sigma\eta_{fX} > c_I$.

Game 2's material and social payoffs for both players (from (A5-4) and A5-4') depend on the expected proportions of groups X and Y who choose to fight (η_{fJ}). Expected values for η_{fJ} allow Group-X extremists to deduce expected values for X's relative share of fighting resources (M_{XY}), group X's probability of success (ρ_{sX}), and the social payoff to fighting ($\sigma\eta_{fX}$).

Furthermore, Chapter 3's Figure 4 (also see Appendix 4) can model conformity dynamics that influence η_{fX}. Here, Chapter 3's discussion of preference falsification regarding support for the status quo and Chapter 5's discussion of possible *unraveling preference falsification*—as one of two paths through which IEs may motivate aggression—apply. As stated, Game 2's outcome depends on within-group expectations of the proportion who choose fight (η_{fJ}). Interpreting Figure 4's Obey strategy as following an IE or extremist directive to fight and, for simplicity, abstracting from member heterogeneity with respect to identity singularity, a revised version Figure 4—this time with nonlinear functions—could show a multi-player assurance game that establishes expected values for η_{fX} (or η_{fY}).

Chapter 5's second path through which IEs motivate aggression, that is, *altered understandings of the difficulty of challenging* the status quo also applies. Here, stage2 incidents increase common perceptions of status-quo vulnerability, reflected in diminished expected values for values for F_Y and F_G in equation (A5-2). Accordingly, group-X player expectations concerning the probability of successful seizure of

contested resource (R_T), equation (A5-2)'s ρ_{sX} increase for any given η_{fX}. In terms of Figure 4, the 'Obey' curve (representing following IE and extremists in fighting) shifts left, as does the 'Disobey' curve. Both shifts lower the critical mass needed to motivate fighting (η_{fX}*), lowering the participation threshold for generating an expectation of widespread group-X fighting. The X's collective-action problem of challenging the R_T status quo correspondingly diminishes.

A more detailed model could thus allow an intermediate step in Game 2, whereby after the Extremist's move, the outcome of a pertinent version of Figure 4 influences variables M_{XY}, ρ_{sX}, and η_{fX} in (A5-2, A5-4, and A5-4'). An additional condition for (Incident, Fight) would then apply:

v. If, under condition iii., $\theta_i M_{XY} R_T + \sigma\eta_{fX} > c_f$, members of X engage in a revised Figure 3 game. If that outcome generates $\eta_{fX} > \eta_{fX}$*, members expect others to fight. The expected member material and social payoffs, $\theta_i M_{XY} R_T + \sigma\eta_{fX}$, approach their values at the maximum Nash equilibrium value for η_{fX} (η_{fXmax}).

vi. With these adjustments, with backward induction, extremist conducts a second incident only if $\theta_i M_{XY} R_T + \sigma\eta_{fXmax} > c_I$.

Under these conditions, IEs and extremists have resolved the collective-action problem of challenging the status quo arrangements regarding allocation of access to contestable resource R_T. New social dilemmas, such as undue group-X, political-economic influence—a new 'resource curse,' likely follow.

More complicated models could specify functions for each payoff variable that add detail to or go beyond the relationships shown here. A series of potentially testable hypotheses (assuming sufficient data) could then follow.

Transition to Stage 3 Fighting

Assume that Game 2 concludes with an (Invest, Fight) outcome. This outcome, summed across members, along with underlying determinates of social categories (C) (from (10), and social cleavage (Θ_{XY}; from 11), η_{fJ} (from relevant Figure 4) and corresponding values for M_{XY} and ρ_{sX} in (A5-2), Υ_{iX} (from A5-3), and finally payoffs $\omega_{iX}(f_{iX})$ and ω_{ExtX} from (A5-4) and (A5-4') determine group X's resources devoted to Fighting (F_X). We have:

(A5-5) $F_X = F_X(\Gamma_{XY}, \Theta_{XY}, \eta_{fX}, M_{XY}, \Upsilon_{iX}, \omega_{iX}, \omega_{ExtX})$.

On this basis, the intensity of fighting over contested resource R_T depends on both groups' devition of resources to fighting as well as underlying singularity and social cleavage (which independent influence on items such as group membership, as well as direct influence on fighting: We have Chapter 5's conflict equation:

(12) $\Omega_{XY} = \Omega_{XY}(\Gamma_{XY}, \Theta_{XY}, F_X(\cdot), F_Y(\cdot))$.

Underlying detail, group affiliation:

Here is a slight elaboration on group affiliation decisions. Recall that Chapter 4 argued that the discretionary element of group affiliation (affiliation is partially socially imposed) depends on the following three factors (Shayo, 2009; Sambanis and Shayo, 2013):

1. A *material factor*: the X–Y difference in anticipated net material benefits.

2. A *cognitive factor*: assessment of the social distance between an individual and the perceived prototypical members of X and Y.

3. An *affective factor*: a comparison of X vs. Y status reflected in relative payoffs.

Equation A5-6 illustrates individual *i*'s payoff to affiliating with group X instead of Y.

(A5-6) $\omega_{iX} = \alpha\Omega(C_t, \Psi_t, S_{kt-1}, Q) + \beta\pi_{iXY}(S_k)) + \gamma(d^2_{iX} - d^2_{iY}) + \delta\Phi_{XY}(R_X - R_Y)$

Here, ω_{iX} = *i*'s net payoff form affiliating with X. The first argument, Ω, represents exogenous historical/cultural bakcground—a function of social categories C_t and norms Ψ_t (both outcomes of history); prior strategies of all involved (S_{kt-1}), and external influences Q. The next three arguments represent individual *i's* discretion. The *material factor* ($\pi_{iXY} \equiv \pi_{iX} - \pi_{iY}$) is the anticipated) X-Y difference in net material benefits which depend on strategies (S_k). The *cognitive factor*, ($d^2_{iX} - d^2_{iY}$), is the Euclidian distance between *i's* exogenous characteristics, ε_i (a weighted vector) and perceived characteristics of prototypical members of X and Y (ε_X, ε_Y). Finally, the *affective factor*, Φ_{XY}, represents the X vs. Y status comparison (Φ_{XY}), which depends on relative resource access ($R_X - R_Y$), such as lootable point-source resources (oil, diamonds) and positions of power. α, β, γ, and δ are parameters, which show the weights of the four terms. These may be quasi-parameters subject to longer-term influence from related variables (such as singularity).

Group affiliations like individual to group social identities. As shown in Chapter 4's equations (7) and (8), and the accompanying text, these terms coevolve (see that discussion).

Now, returning to the intensity of conflict represented in equation (12), identity singularity Γ_{XY}, influences cognition regarding social distance ($\gamma(d^2_{iX} - d^2_{iY})$). Singularity also frames interpretations of relative status in the *affective factor* ($\delta\Phi_{XY}(R_X - R_Y)$) and even anticipated direct material payoffs $\beta\pi_{iXY}$.

Social cleavage (Θ_{XY}) exerts similar effects.